

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular & Supplementary Summer 2024 Course: TY B. Tech. Branch :Mechanical Engineering Semester :VI Subject Code & Name: BTMPE603A IC Engines Max Marks: 60 Date: 19/06/2024 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Classify the internal combustion engine.	CO1	6
B)	Explain valve timing diagram for two stroke engine.	CO2	6
C)	How actual cycle deviates from air standard cycle.	CO2	6
Q.2	Solve Any Two of the following.		12
A)	Explain stages of combustion in CI Engines.	CO2	6
B)	Define octane number. Explain how HUCR is related to it.	CO5	6
C)	What are the various types of combustion chambers used in SI engines? Explain any two with neat sketch.	CO3	6
Q. 3	Solve Any Two of the following.		12
A)	Describe battery ignition system with neat sketch.	CO3	6
B)	Explain pressure lubrication system with neat sketch.	CO1	6
C)	What is the necessity of governing of an IC engine. Explain mechanical governor with neat sketch.	CO2	6
Q.4	Solve Any Two of the following.		12
A)	What is the effect of supercharging in SI and CI engines.	CO2	6
B)	A six-cylinder, gasoline engine operates on four stroke cycle. The bore of each cylinder is 70 mm and the stroke 100 mm. The clearance volume per cylinder is 67 CC. At a speed 3960 rpm, the fuel consumption is 19.5 kg/hr. and the torque developed is 140 Nm Calculate i) the brake power ii) the brake mean effective pressure iii) brake thermal efficiency if the calorific value of fuel is 43000	CO5	6

	KJ/kg and iv) the relative efficiency on brake power basis assuming the engine works on constant volume cycle. $\gamma = 1.4$		
C)	Explain exhaust gas recirculation in engine emission control.	CO2	6
Q. 5	Solve Any Two of the following.		12
A)	Why there is need of alternative fuel? list out different alternative fuels.	CO4	6
B)	Explain the concept of fuel cell technology.	CO2	6
C)	Explain advantages and drawbacks of electric and hybrid vehicles.	CO2	6
	*** End ***		